Fiscal Year:	FY 2018	**Task Last Updated:**	FY 04/16/2019
PI Name:	Smith, Scott M Ph.D.		
Project Title:	Biochemical Profile: Homozygous Twin Control for a 12 Month Space Flight Exposure		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	scott.m.smith@nasa.gov	**Fax:**	FY 281-483-2888
PI Organization Type:	NASA CENTER	**Phone:**	281-483-7204
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Biomedical Research and Environmental Sciences Division/SK3		
PI Address 2:	2101 NASA Pkwy		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	07/01/2014	**End Date:**	04/30/2018
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Charles, John	**Contact Phone:**	281-483-7224
Contact Email:	jbcharle@gmail.com		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 4/30/2018 to align with other TWINS studies (Ed., 12/14/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zwart, Sara Ph.D. (University of Texas Medical Branch) Heer, Martina Ph.D. (University of Bonn, Germany)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As long-duration spaceflights continue on the International Space Station (ISS), and the operational suite of countermeasures is modified, the food system is updated, and the duration of missions lengthens, it will be important to evaluate and monitor a broad set of biomarkers for key physiological systems. The Nutritional Status Assessment Supplemental Medical Objective (aka “Nutrition SMO”) was initiated in 2006. It consisted of blood and urine collections twice before, up to 5 times during, and twice after long-duration space flight to measure kinetics of changes in nutritional status and provided a space biochemical database that has already yielded many significant findings with clinical, operational, and scientific implications. Now that the Nutrition SMO has ended, the database is being extended through the newly selected Biochemical Profile project for future long-duration crewmembers, including the flight subject of the TWINS mission. We propose here to extend the Biochemical Profile to the identical twin as a ground-based control. By doing this, there will be continuity of data collection and analysis protocols, and it will ensure continued scientific integrity of the database, and will support many of the studies being selected in this call for proposals. These findings will allow better understanding of human adaptation to flight, and of countermeasures effectiveness on orbit.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To paraphrase Dr. John Charles -- all NASA research has terrestrial implications of one form or another. This study, and the larger Twins Study overall will advance NASA research into areas never considered until this effort. The One Year Mission begins to explore space flights of durations longer than the US has ever considered. The results of this study will allow for a detailed understanding of the effects of long-duration space flight on human physiology and a broad biochemical analysis, including estimates of changes in bone, cardiovascular, endocrine, and immune systems and more. Understanding these changes during long-duration flight inform our understanding of terrestrial human physiology.
Task Progress:	Study is now complete, and the data have been published. Key findings from our perspective include that the results document that nutrition is more important than many want to give credit. Telomere changes can be ascribed to increasing folate status and body mass loss. Incidence of ocular (space flight-associated neuro-ocular syndrome (SANS)) issues fits previous research documenting genetic and B-vitamin status predisposing some individuals to SANS. This is a great first study of missions longer than the usual 4-6 months. We need to study more crews to get a better understanding of group (and subgroup) effects of these types of missions. Unfortunately, the Biochemical Profile study is no longer active.
Bibliography Type:	Description: (Last Updated: 01/19/2026)
Articles in Peer-reviewed Journals	Garrett-Bakelman FE, Darshi M, Green SJ, Gur RC, Lin L, Macias BR, McKenna MJ, Meydan C, Mishra T, Nasrini J, Piening BD, Rizzardi LF, Sharma K, Siamwala JH, Taylor L, Vitaterna MH, Afkarian M, Afshinnekoo E, Ahadi S, Ambati A, Arya M, Bezdan D, Callahan CM, Chen S, Choi AMK, Chlipala GE, Contrepois K, Covington M, Crucian BE, De Vivo I, Dinges DF, Ebert DJ, Feinberg JI, Gandara JA, George KA, Goutsias J, Grills GS, Hargens AR, Heer M, Hillary RP, Hoofnagle AN, Hook VYH, Jenkinson G, Jiang P, Keshavarzian A, Laurie SS, Lee-McMullen B, Lumpkins SB, MacKay M, Maienschein-Cline MG, Melnick AM, Moore TM, Nakahira K, Patel HH, Pietrzyk R, Rao V, Saito R, Salins DN, Schilling JM, Sears DD, Sheridan CK, Stenger MB, Tryggvadottir R, Urban AE, Vaisar T, Van Espen B, Zhang J, Ziegler MG, Zwart SR, Charles JB, Kundrot CE, Scott GBI, Bailey SM, Basner M, Feinberg AP, Lee SMC, Mason CE, Mignot E, Rana BK, Smith SM, Snyder MP, Turek FW. "The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight." 2019 Apr 12;364(6436):eaau8650. https://science.sciencemag.org/content/364/6436/eaau8650 ; PubMed PMID: 30975860 , Apr-2019
Articles in Peer-reviewed Journals	Sides MB, Johnston SL 3rd, Sirek A, Lee PH, Blue RS, Antonsen EL, Basner M, Douglas GL, Epstein A, Flynn-Evans EE, Gallagher MB, Hayes J, Lee SMC, Lockley SW, Monseur B, Nelson NG, Sargsyan A, Smith SM, Stenger MB, Stepanek J, Zwart SR. "Bellagio II Report: Terrestrial applications of space medicine research." Aerosp Med Hum Perform. 2021 Aug 1;92(8):650-69. https://doi.org/10.3357/AMHP.5843.2021 ; PMID: 34503618 , Aug-2021
Articles in Peer-reviewed Journals	Walle M, Gabel L, Whittier DE, Liphardt AM, Hulme PA, Heer M, Zwart SR, Smith SM, Sibonga JD, Boyd SK. "Tracking of spaceflight-induced bone remodeling reveals a limited time frame for recovery of resorption sites in humans." Sci Adv. 2024 Dec 20;10(51):eadq3632. http://dx.doi.org/10.1126/sciadv.adq3632 ; PMID: 39705358; PMCID: PMC11661419 , Dec-2024
Journal/Magazine covers	Garrett-Bakelman FE, Darshi M, Green SJ, Gur RC, Lin L, Macias BR, McKenna MJ, Meydan C, Mishra T, Nasrini J, Piening BD, Rizzardi LF, Sharma K, Siamwala JH, Taylor L, Vitaterna MH, Afkarian M, Afshinnekoo E, Ahadi S, Ambati A, Arya M, Bezdan D, Callahan CM, Chen S, Choi AMK, Chlipala GE, Contrepois K, Covington M, Crucian BE, De Vivo I, Dinges DF, Ebert DJ, Feinberg JI, Gandara JA, George KA, Goutsias J, Grills GS, Hargens AR, Heer M, Hillary RP, Hoofnagle AN, Hook VYH, Jenkinson G, Jiang P, Keshavarzian A, Laurie SS, Lee-McMullen B, Lumpkins SB, MacKay M, Maienschein-Cline MG, Melnick AM, Moore TM, Nakahira K, Patel HH, Pietrzyk R, Rao V, Saito R, Salins DN, Schilling JM, Sears DD, Sheridan CK, Stenger MB, Tryggvadottir R, Urban AE, Vaisar T, Van Espen B, Zhang J, Ziegler MG, Zwart SR, Charles JB, Kundrot CE, Scott GBI, Bailey SM, Basner M, Feinberg AP, Lee SMC, Mason CE, Mignot E, Rana BK, Smith SM, Snyder MP, Turek FW. "Cover of Science for 'The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight.' " Science. 2019 Apr 12;364(6436):eaau8650. See cover image on right in link: https://science.sciencemag.org/content/364/6436/eaau8650 , Apr-2019